SkillBazaar

Project Team

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Chapter 1

Introduction

Skill Bazaar focuses on revolutionizing the current blue collar job marketplace by integrating automation and AI to build a platform where illiterate people can easily sell their services. It will tend to decrease unemployment caused by informal networks of word-to-mouth that limits job opportunities.

1.1 Problem Statement

We are developing this mobile application as a bilingual platform to empower the illiterate and poor blue collar job workers like Gardeners, plumbers, electricians, daily wage laborers and women in handicraft etc. There is unavailability of an online platform which will connect blue collar job seekers with employers directly to help them find jobs efficiently which tends to reduce unemployment. Many people remain in poverty because of lack of opportunities and under utilization of their skills. The informal network of work-to-mouth is limiting opportunities and is reducing potential income of the daily wage workers. Sometimes, they are unaware of the new job opportunities as there is no such marketplace like Fiverr, Upwork etc for Blue collar workers. Due to their inability to read and write English and Urdu, sometimes they are uninformed or they are paid way less than they deserve because of their innocence and on the other hand when people call some blue collar worker to their house, they struggle to find a reliable and trustworthy person since there is lack of transparency. To solve all these problems we need a streamlined blue collar job marketplace that would help the workers find jobs and communicate more efficiently and transparently.

1.2 Scope

The main scope of our project is to automate the whole process of the blue collar job marketplace. We are targeting people and businesses in Pakistan looking for skilled workers or small jobs. Our product's key features include Automated Profile building using cnic and Voice Input, Job Postings, Bookings, Ratings, Multilingual support, voice based conversational bot, Geo-Location Services, Voice based Searches and User Analytics. This means that the daily wage workers can get their profiles created and find a suitable job with their minimal input.

1.3 Modules

Given the current scope, our products will be able to implement the following modules.

1.3.1 Automated Profile Creation Module (OCR & Text Input)

The first Module is basically the building block of our product that partially automates the profile for our users.

1. Uses OCR for extracting the data from CNIC.
2. Text based input is taken from the user for profile creation.
3. It supports both English and Urdu input.
4. Profile data validation.

1.3.2 Application Management Module

In this module the application is being managed from the admin side.

1. Job postings and updates.
2. Job search by filters (location, price, service type).
3. Bookings and confirmations.
4. In-app messaging for the users.
5. Rating and feedback system.

1.3.3 Conversational Bot Module (ASR Technology)

Our third module is generating a conversational bot for the ease of our users.

1. Voice generation and understanding.
2. Voice-based assistance for navigating the app.
3. ASR technology for job posting, searching, and profile management.

1.3.4 AI Matching Algorithm Module

Our fourth module is the most important which would match the jobs with services.

1. Matches users based on location, price, job type, and services.
2. Considers user preferences and availability.
3. Dynamic adjustment for demand and supply in different regions.
4. Personalized job or service recommendations.

1.4 User Classes and Characteristics

These are the major user classes that will anticipate with our product and these are their pertinent characteristics.

User class	Description
Blue collar worker	A blue collar worker is a daily wage worker who is looking for a job or work and is offering labor internsive services. The worker maybe electrician, plumber, carpenter, a women in handicraft etc. He/she is likely to have limited literacy, is to some extent unable to read and write and will prefer voice input over text. The worker doesn't have mainly mobile usage experience.
Employers	The employers can be individuals, businesses, organizations or a company in need of any kind of worker mainly focusing on blue collar workers. People looking for skilled workers.
Job Seekers	The job seekers are skilled people, who want to sell their services but they are not literate enough to find jobs on upwork, fiver, daraz, uber or instagram so they can easily find their customers using our platform.
Non-Literate Users	As the most users maybe from internal areas of Pakistan which means that they can not read and write in English so we have multilingual support and voice support for them so that they can easily communicate, look or give jobs and get their work done and earn money.

Example: User Classes and Characteristics for Skill Bazaar

Chapter 2

SRS

We are developing an application which will automate the profiles and job findings for the blue collar job workers. This system will incorporate automated profile creation, job matching and assessment. We are providing urdu voice based conversational bot for our users who are deprived of basic education. Our goal is to promote skill utilisation and to make job finding easy for blue collar job holders and create a transparency between the employer and the employee.

2.1 Lists of Features

These are the list of features that our product will have;

1. Profile creation
2. Multilingual Support
3. Conversational bot (voice based)
4. Service/Job posting
5. Service/Job matching
6. Profile matching
7. Profile management
8. Geo-location service

2.2 Use-case/Event Response Table/Story-boarding

The selection of the requirement gathering technique(s) will depend on the type of project. For instance,

2.2.1 Use case

Following the basic use-case diagram followed by the detailed use-cases tables.

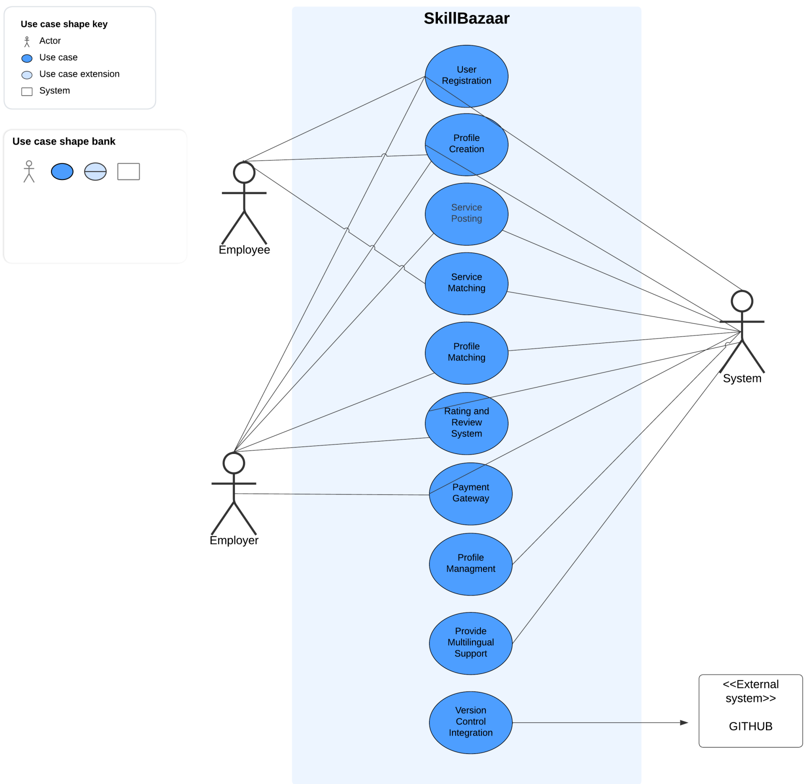

Figure 2.1: Use-Case Diagram

2.2.2 Event- response table

This is the event-response table in which there are event is from the customer side where as our system generates response based on the events.

Event	Response
Customer	System
User Provides CNIC Image	Backend fetches the text from the CNIC using OCR and initiates automated profile creation.
User inputs text	Backend takes the inormation linking it with the database and completing the profile creation along with CNIC information.
User inputs voice to create profile	Backend uses voice recognition to transcribe input and links it to CNIC information for completion of profile.
User posts job request	Backend stores job data, runs AI matching algorithm and sends notifications to the relevant workers with in geographic location.
Worker submits a job application	Backend notifies the neighbouring businesses that are interested in the work, updates the user profile and tracks the status of the application i.e valid or invalid.
User requests voice based job search	Backend processes voice input, performs search, and returns relevant job postings based on location and skills.
User gives a rating after a job	Backend updates the worker's rating, adjusts worker ranking in the marketplace, and sends feedback to the worker.
User provides new information	Backend updates user information and refines job or worker search results, and sends local job alerts according to new information.
Worker accepts job request	Backend confirms booking, updates job status, and notifies the business.
Business user requests analytics	Backend processes user data to generate reports on worker performance, job fulfillment rates, etc.
Voice-based chatbot receives a query	Backend processes the query, interacts with the database and responds with contextual information.
User switches app language	Backend updates user location, refines job or worker search results, and sends local job alerts.
User performs voice-based search	Backend uses the voice understanding and ASR techniques to conducts a search in the database, and returns results in the desired language. .

2.2.3 Story boarding

:

Following are the first three pages of our mobile application. The detailed version is in the chapter 5.

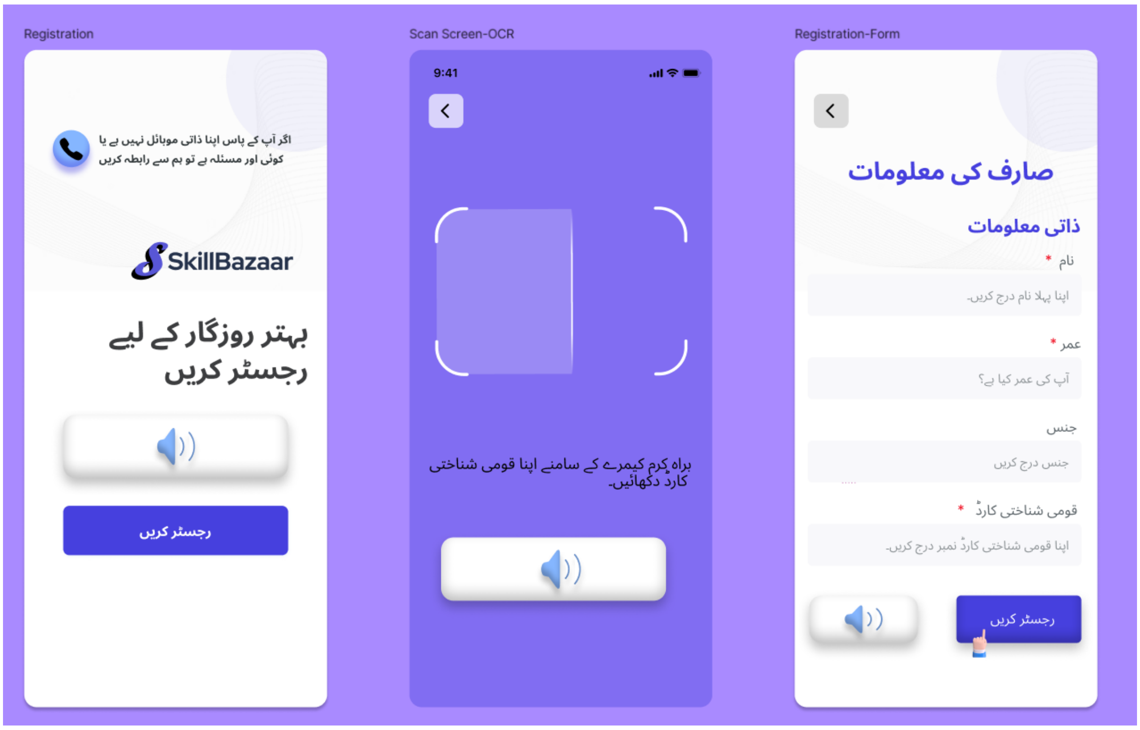

Figure 2.2: Story Boarding

2.3 Functional Requirements

This section describes the functional requirements of the Skill Bazaar mobile application system, arranged by key modules.

2.3.1 Profile creation

Using OCR and ASR technology to scan out Urdu/English text from the images and scanning voice input to automate the profile for the users respectively.

1. English Text detection from CNIC.
2. Urdu Text detection from CNIC.

3. Speech recognition and understanding.

4. Extraction of important information from the voice input.

5. The information retrieved will be stored into the database.

2.3.2 Multilingual Support

The will be multi-lingual support for our users as the main audience for our application is blue collar job holders who can not read and write English in particular.

1. English.

2. Urdu.

2.3.3 Conversational bot

The application will allow the users to communicate via their voice as the targeted people may or may not be literate enough.

1. Speech recognition and understanding.

2. Urdu voice based conversation and correct interpretation.

3. Quick and easy to understand replies.

2.3.4 Service/Job posting and Matching

Another functional requirement is the service and job posting and matching.

1. Service is posted by the workers who aim to secure a job.

2. Jobs will be posted by the people looking for any kind of blue collar worker.

3. The application will manage both type of the postings.

4. An AI algorithm will match both the services and jobs according to the user or worker's preference of work.

5. Geo-location will also be matched along with other preferences.

6. There is a voice based conversational bot for this module as well.

2.3.5 Profile matching and maintenance

The profile of the users and workers will also be matched and as the application is solely built on automating the profile and job seeking so we will focus on following;

1. Profile matching based on rating, reviews, geo-location and other preferences etc.
2. Profile matching can also allow voice input.
3. System will manage the profile for the workers in case they want to add any new information.

2.4 Non-Functional Requirements

This section specifies nonfunctional requirements. These quality requirements should be specific, quantitative, and verifiable. The following are some examples of documenting guidelines.

2.4.1 Reliability

Below is the requirements of how often the system can fail and how quickly can it recover.
REL-1: The system shall have a **Mean Time Between Failures (MTBF)** of at least 500 hours for the user profile creation and job posting modules.

REL-2: In the event of failure, the system shall recover and resume normal operations within **2 minutes** after detecting the failure.

REL-3: The system shall implement error detection mechanisms for failed job postings or profile updates and notify the user within **5 seconds** of the error.

REL-4: The system shall store incomplete user sessions and restore them upon next login if a failure occurs during user interaction.

2.4.2 Usability

The usability requirements specified here will help the user interface designer create the optimum user experience. Example:

USE-1: The system shall allow a user to create a profile using CNIC and text/voice input within **3 interactions**.

USE-2: The system shall support **multilingual navigation** with options for 2 major languages (Urdu, English) and allow users to switch between languages within **1 click**.

USE-3: The system shall allow users to post a job in under **1 minute** by filling a simple form or using a voice command.

USE-4: The system shall ensure all screens, including job postings and ratings, are **screen-reader compatible** to meet accessibility requirements.

USE-5: The system shall provide clear, on-screen guidance and **error recovery prompts** in the event of invalid inputs during profile creation or job posting.

2.4.3 Performance

Following are the performance goals we aim to achieve for this system.

PER-1: **95%** of all screens, including profile creation, job searches, and bookings, shall load within **3 seconds** under a **4G network** connection.

PER-2: The system shall process job search queries and return relevant results within **2 seconds** for workers within a **50 km** geographic radius.

PER-3: The system shall extract and process text from images using OCR within **4 seconds** for images with a resolution of **1920x1080** or lower.

PER-4: The system shall support up to **10,000 concurrent users** during peak hours without performance degradation.

2.4.4 Security

The security measure are as follow;

SEC-1: The system is made for less educated people hence there is one time login for them but one person can operate his/her account on **one device** only.

SEC-2: The system shall encrypt all sensitive data, including CNIC details and job information, using **AES-256** encryption.

SEC-3: The system shall ensure that all uploaded documents and images are scanned for **malware or virus threats** before being stored in the database.

2.5 Domain Model

This is the representation of our domain model which clearly shows how the application is linked, describing the hierarchy.

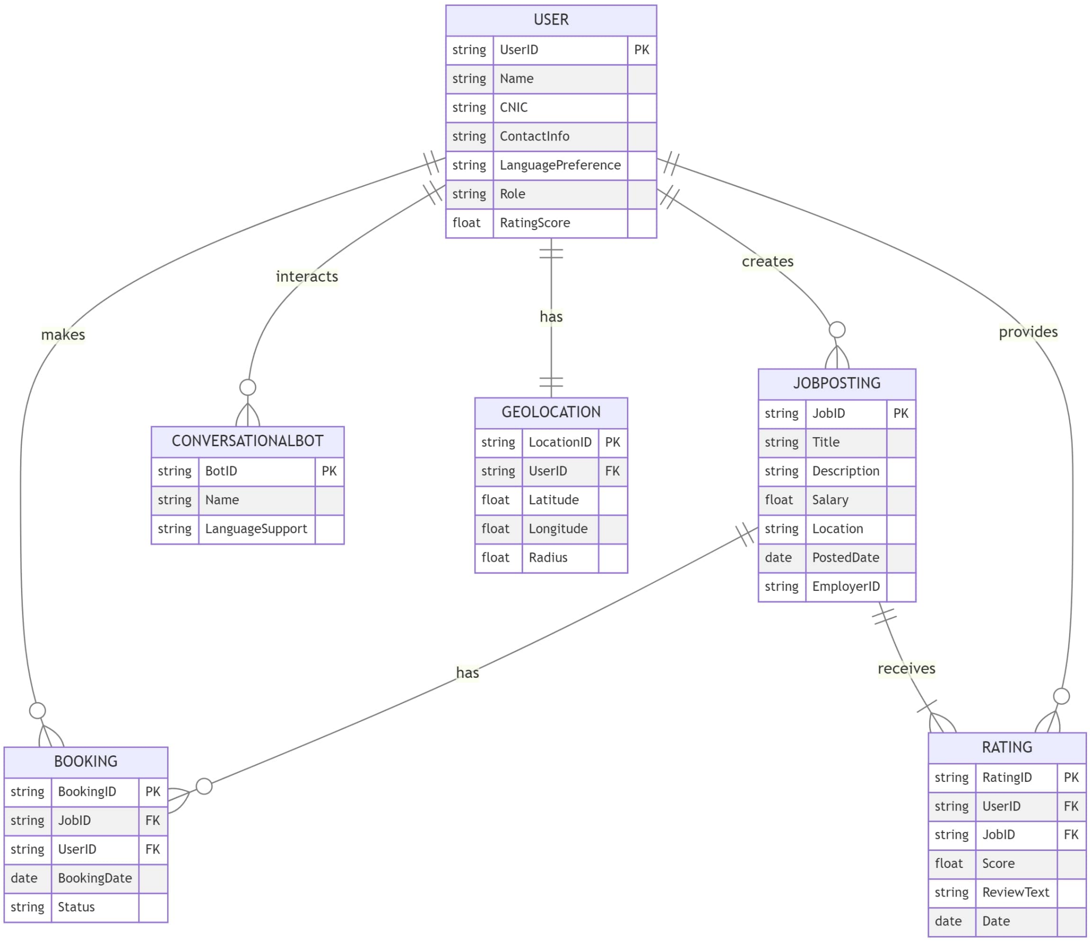

Figure 2.3: Domain Model Diagram

Chapter 3

Use Case

3.1 High-Level Use Cases

The main goal of our application is to automate the profiles and the job finding for the blue collar people. The use cases are mainly related to ease of use and integrating AI to make job finding process easy for the daily wage workers allowing them to sell their services using less input. Following are the high level use cases of the product, a basic over view with the help of tables that would make it easy to understand:

1. User Authentication
2. Profile Creation
3. Multilingual Support
4. Service/Job Posting
5. Job/Service Matching
6. Conversational Bot
7. Payment Gateway
8. Ratings/Reviews

Use Case	User/Account Authentication and Real-Time Image Processing with OCR
Actors	User, System
Type	primary
Description	The user will be able to perform a real-time scan of an image, and system will extract information from the image using OCR (Optical Character Recognition), and register an account based on the extracted data.

Table 3.1: Use Case 1

Use Case	Create user profile through voice/Text input.
Actors	User System (Voice Recognition and Profile Creation)
Type	primary
Description	The user will be able to create a profile in an application using voice input. The system captures spoken details such as name, email, address, and other necessary information, converts them into text, and automatically fills out the profile creation form.

Table 3.2: Use Case 2

Use Case	Multilingual Support for Application
Actors	System (with multilingual capabilities)
Type	primary
Description	Implementation of multilingual support within an application, allowing users to interact with the system in their preferred language (Urdu & English). The system dynamically translates content, ensuring a seamless user experience across different languages (Urdu & English).

Table 3.3: Use Case 3

Use Case	Voice/Text-Based Service Posting
Actors	User (Service Provider) System (Voice Recognition or Text Understanding and Service Posting)
Type	Secondary
Description	Allowing users to post services through voice commands within an application. The system uses voice recognition to convert spoken input into text and automates the service posting process based on the user's input.

Table 3.4: Use Case 4

Use Case	Job Matching Similar Services after Service Posting and Profile Creation
Actors	User (Service Provider) Job Poster (Employer) System (Job Matching and Recommendation Model)
Type	Secondary
Description	Automated matching of jobs with relevant services and user profiles after a service is posted and a profile is created. The system compares job requirements with service offers to find suitable matches.

Table 3.5: Use Case 5

Use Case	Voice-Enabled Chatbot Assistance with Voice Recognition and Response
Actors	System (with Voice Recognition/Understanding and Response)
Type	Secondary
Description	Providing chatbot assistance through voice input, users will be able to interact with the system by speaking instead of typing. The system processes the voice input, responds in real-time, and assists the user with queries.

Table 3.6: Use Case 6

Use Case	Payment Gateway Integration
Actors	Customer System
Type	Secondary
Description	After job completion user will be able to make a payment through an integrated payment gateway system. The system handles secure payment processing, validates transactions, and confirms payment status.

Table 3.7: Use Case 8

Use Case	Rating Review System
Actors	Customer
Type	Secondary
Description	Providing chatbot assistance through voice input, users will be able to interact with the system by speaking instead of typing. The system processes the voice input, responds in real-time, and assists the user with queries.

Table 3.8: Use Case 9

3.2 Expanded Use Cases

Following are the expanded use cases of the product, a broader over view with the help of tables that would make it easy to understand:

1. User Authentication
2. Profile Creation
3. Multilingual Support
4. Service/Job Posting
5. Job/Service Matching
6. Conversational Bot
7. Payment Gateway
8. Ratings/Reviews

Use Case ID:	UC01
Use Case Name:	User/Account Authentication and Real-Time Image Processing with OCR
Scope:	Application for user registration and real-time image scanning
Level:	System
Primary Actor:	Customer, System
Stake Holders and Interests	- **Customer:** Wants their account securely authenticated and relevant information accurately extracted from images (e.g., CNIC, ID card). - **System:** Ensures the accuracy of OCR processing and secure authentication of the user.
Main Success Scenario	**Actor Actions:** - User enters their login credentials. - User is prompted to scan/upload an image (e.g., ID card) using OCR. **System Response:** - System verifies the credentials and authenticates the user. - The system processes the image, extracts the necessary details using OCR, and matches them with the profile information.
Extensions	**Scenario A:** The user enters incorrect credentials. The system notifies the user of incorrect credentials and prompt re-entry. **Scenario B:** The system fails to extract correct information using OCR. The user is notified to re-scan the image or manually enter the details.
Pre-Conditions	- The user must have an account or valid credentials. - Internet connectivity must be available.
Post-Conditions	- User is authenticated and logged in. - OCR successfully extracts and updates user profile data.

Table 3.9: User Authentication

Use Case ID:	UC02
Use Case Name:	Create User Profile through Voice Input
Scope:	Voice-based user profile creation
Level:	System
Primary Actor:	System
Stake Holders and Interests	- **User:** Wants an easy and quick way to create a profile via voice commands. - **System:** Wants to ensure voice recognition is accurate and data is correctly processed.
Main Success Scenario	**Actor Actions:** - User selects the voice input option for profile creation. - User provides information via voice. - User reviews the auto-filled profile and submits it. **System Response:** - System prompts user for basic details (e.g., name, email). - System converts voice input into text and fills in the profile fields. - System saves the profile and confirms successful creation.
Extensions	**Scenario A:** - The system misinterprets the voice input. - The system requests confirmation or prompts re-entry for the misinterpreted fields. **Scenario B:** - The user provides incomplete information. - The system prompts the user for missing details before saving.
Pre-Conditions	- User has access to a microphone-enabled device. - The system has voice recognition capabilities.
Post-Conditions	- User profile is successfully created with voice input.

Table 3.10: Profile Creation

Use Case ID:	UC03
Use Case Name:	Multilingual Support for Application
Scope:	Application offering services in multiple languages
Level:	System
Primary Actor:	System
Stake Holders and Interests	- **User:** Wants the ability to use the app in their preferred language. - **System**: Must accurately translate UI and content based on user language preferences.
Main Success Scenario	**Actor action:** - User selects their preferred language from the settings. - The user continues to interact with the app in their chosen language. **System Response:** - System updates the UI and content to reflect the selected language.
Extensions	**Scenario A:** - System fails to fully translate a section of the UI. - The untranslated parts revert to the default language, and a notification is sent for translation errors.
Pre-Conditions	- The application must support multiple languages.
Post-Conditions	- The user is able to use the app in their selected language.

Table 3.11: Multilingual Support

Use Case ID:	UC04
Use Case Name:	Voice-Based Service Posting
Scope:	Application with voice-enabled service posting
Level:	System
Primary Actor:	System, Customer
Stake Holders and Interests	- **Customer**: Wants to quickly post services via voice input. - **System:** Ensures accurate voice-to-text conversion and successful posting of services.
Main Success Scenario	**Actor action:** - User chooses to post a service using voice input. - User describes the service via voice (e.g., type, price, availability). - User reviews the form and submits the service. **System Response:** - System prompts the user for service details. - System processes voice input, converts it to text, and fills out the service posting form. - System successfully posts the service.
Extensions	**Scenario A:** - Voice input is not recognized accurately. - The system requests the user to repeat the description or correct the form manually
Pre-Conditions	- Voice recognition must be enabled.
Post-Conditions	- The service is successfully posted with details provided through voice input

Table 3.12: Service/Job Posting

Use Case ID:	UC05
Use Case Name:	Matches Jobs with Similar Services after Service Posting and Profile Creation
Scope:	Job matching based on service postings and profile information
Level:	System
Primary Actor:	System
Stake Holders and Interests	- **Customer:** Wants relevant job recommendations based on posted services and profiles. - **System:** Ensures accurate matching between job postings and service providers**.**
Main Success Scenario	**Actor action:** - User User creates a profile or posts a service. - User is notified of the job matches. **System Response:** - System analyzes job postings for matches with the service or profile. - System generates a list of matching jobs.
Extensions	**Scenario A:** - No relevant job matches are found. - The system informs the user and continues monitoring new postings for potential matches.
Pre-Conditions	- User profile or service must be created/posted. - The system must have access to job postings
Post-Conditions	- The user is provided with job matches based on their services and profile.

Table 3.13: Job Matching

Use Case ID:	UC06
Use Case Name:	Voice-Enabled Chatbot Assistance with Voice Recognition and Response
Scope:	Voice-enabled chatbot for customer support
Level:	System, Customer
Primary Actor:	System, Customer
Stake Holders and Interests	- **Customer:** Wants voice-enabled assistance for inquiries or issues. - **System:** Processes voice inputs and responds accordingly.
Main Success Scenario	**Actor action:** - User initiates the chatbot and speaks their query. - The conversation continues based on the user's input. **System Response:** - The system processes the voice input using voice recognition. - The system generates a relevant response, which is either spoken or displayed. The conversation continues based on the user's input and system responses.
Extensions	**Scenario A:** - System does not recognize the user's voice command. - The system asks the user to repeat or offers help with commonly used commands.
Pre-Conditions	- Voice recognition and chatbot functionality must be enabled.
Post-Conditions	- User receives voice-based assistance from the chatbot.

Table 3.14: Conversational Bot

Use Case ID:	UC07
Use Case Name:	Payment Gateway Integration
Scope:	Application integrating a payment gateway for transactions
Level:	System, Customer
Primary Actor:	System, Customer
Stake Holders and Interests	- **Customer:** Wants to complete transactions securely using integrated payment options. - **System:** Ensures secure and reliable payment processing.
Main Success Scenario	**Actor action:** - User selects the option to make a payment - User enters payment details. - User is notified of payment success or failure. **System response:** - System redirects to the payment gateway. - System processes the payment.
Extensions	**Scenario A:** - Payment fails due to insufficient funds. - System notifies the user and allows them to retry with a different method.
Pre-Conditions	- Payment gateway must be integrated with the application.
Post-Conditions	- Payment is processed successfully.

Table 3.15: Payment Gateway

Use Case ID:	UC08
Use Case Name:	Rating & Review System
Scope:	In-app rating and review functionality
Level:	Customer, System
Primary Actor:	Customer, System
Stake Holders and Interests	- **Customer:** Wants to provide feedback on services/products. - **System:** Collects and displays ratings and reviews for transparency.
Main Success Scenario	**Actor action:** - User selects a service or product to rate/review. - User submits their feedback. **System response:** - System prompts the user for a rating and review. - System posts the review and updates the rating.
Extensions	**Scenario A:** - Review contains inappropriate content. - The system flags it for moderation.
Pre-Conditions	- User must have interacted with the service/product.
Post-Conditions	- Rating and review are posted for others to see .

Table 3.16: Rating/Reviews System

Chapter 4

System Overview

SkillBazaar connects blue-collar workers with job opportunities through a simple, multilingual platform designed for ease of use. Many workers and small business employers face challenges with digital tools due to literacy or tech knowledge gaps, but SkillBazaar overcomes these with voice input and OCR technology. Users can easily create profiles, post jobs, and find work with minimal effort. The app uses AI to match workers with jobs based on skills and location, while Geo-location and voice-based conversations enhance accessibility. With its focus on usability and automation, SkillBazaar aims to transform the job market for Pakistan's blue-collar workforce.

4.1 Architectural Design

SkillBazaar's system is designed with a **multi tiered architecture** to ensure scalability, maintainability, and ease of development. The system is decomposed into key subsystems that interact with each other to handle various functionalities like profile creation, job matching, and multilingual support. Below is a high-level description of the modules and their relationships:

1. User Interface (UI) Layer

- **Description:** The front-end of the system, where users interact with the app. It includes voice and text inputs for profile creation, job posting, and searching.
- **Modules:**
  - **Profile Creation UI:** This handles OCR scanning and voice input for profile automation.
  - **Job Posting/Search UI:** It Allows users to post and search for jobs.

    - **Multilingual Support UI:** It provides language selection and voice interactions in Urdu/English.

- **Interaction:** There will be communicates with the application logic through APIs to fetch or send data based on user inputs.

2. Application Logic (Business Layer)

- **Description:** The core functionality is that processes user data, manages profiles, jobs, and matches workers to jobs.
- **Modules:**
    - **Profile Manager:** It handles creation and updates of user profiles using OCR and voice input data.
    - **Job Manager:** This manages job postings by employers and service postings by workers.
    - **Matching Engine:** There will be used AI to match workers with jobs based on skill sets, geo-location, and preferences.
    - **Multilingual Manager:** This will ensure the interactions are in the user's preferred language (Urdu/English).
- **Interaction:** It orchestrates the flow of data between the UI and the backend (data layer), calling the necessary services (AI, geo-location) to execute tasks.

3. Data Layer

- **Description:** This layer manages the persistent storage and retrieval of data, including user profiles, job postings, ratings, and preferences.
- **Modules:**
    - **Database Manager:** It isr esponsible for CRUD operations on user profiles, job postings, and matching history.
    - **Geo-Location Database:**It stores and retrieves location-based data to enable job proximity matching.
    - **Rating & Review Storage:** It manages worker ratings and employer reviews for profile matching.
- **Interaction:** This communicates with the application logic layer, fetching data when requested and updating it as needed.

4. AI & Geo-Location Services (Service Layer)

- **Description:** Provides specialized services to enable job matching and voice-based interaction.
- **Modules:**
  - **OCR/ASR Service:** Processes CNIC images and voice input to extract text and relevant details.
  - **AI Matching Service:** Matches job seekers to job postings using user preferences, skills, ratings, and geo-location.
  - **Geo-Location Service:** Provides proximity-based job matching.
- **Interaction:** Interacts with the business logic layer to provide services when necessary, such as extracting text or matching users to jobs.

5. Conversational Bot (Voice Interaction)

- **Description:** Manages all voice-based communication within the app, facilitating users who may be less comfortable with text input.
- **Modules:**
  - **Voice Input Handler:** Captures and processes voice commands for profile creation, job posting, or search.
  - **Response Generator:** Provides quick and accurate voice replies in Urdu or English based on user interactions.
- **Interaction:** Integrates with the UI and application logic layers to ensure smooth voice interactions and real-time responses.

In this architecture diagram it can be seen that the user is asked to take a picture of his/her CNIC and then OCR will be used for information retrieval then both the voice and text input would be taken from the user based on their preferences and after that the profile will be created, job posting and searching is also voice assisted and there is also a voice based conversational bot for user's ease which will end up in successful job matching that is done using AI matching algorithm.

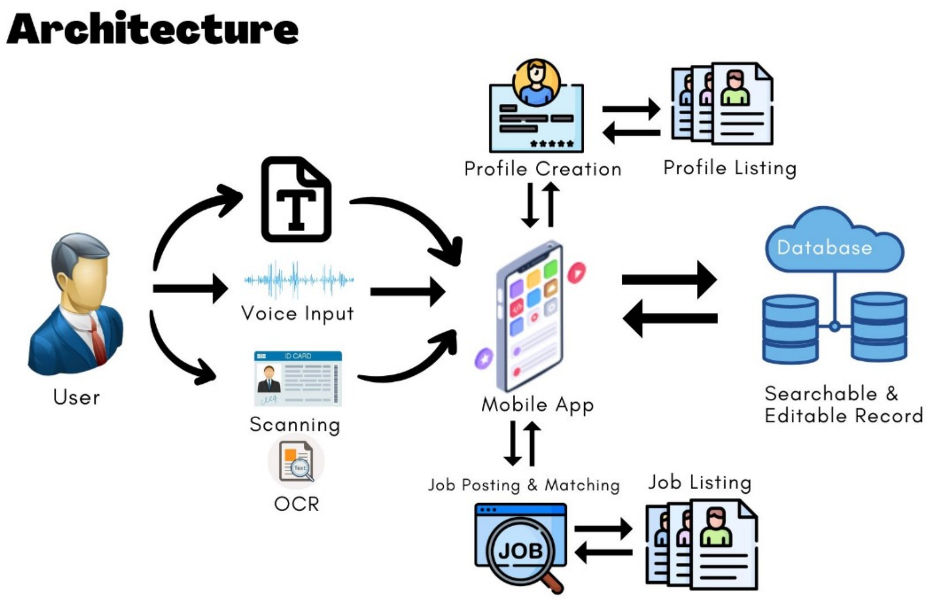

Figure 4.1: Simple Architectural Diagram

4.2 Design Models

Design Models for Object Oriented Development Approach

• Activity Diagram

In this basic activity diagram it can be seen that the user is asked to take a picture of his/her CNIC and then OCR will be used for information retrieval and for more information the user can input using both text and voice, after that the profile will be created, job posting and searching is also voice assisted and there is also a voice based conversational bot for user's ease which will end up in successful job matching that is done using AI matching algorithm.

1. ID scan
2. Text/voice input
3. Profile creation
4. Jobs searching

5. Job matching

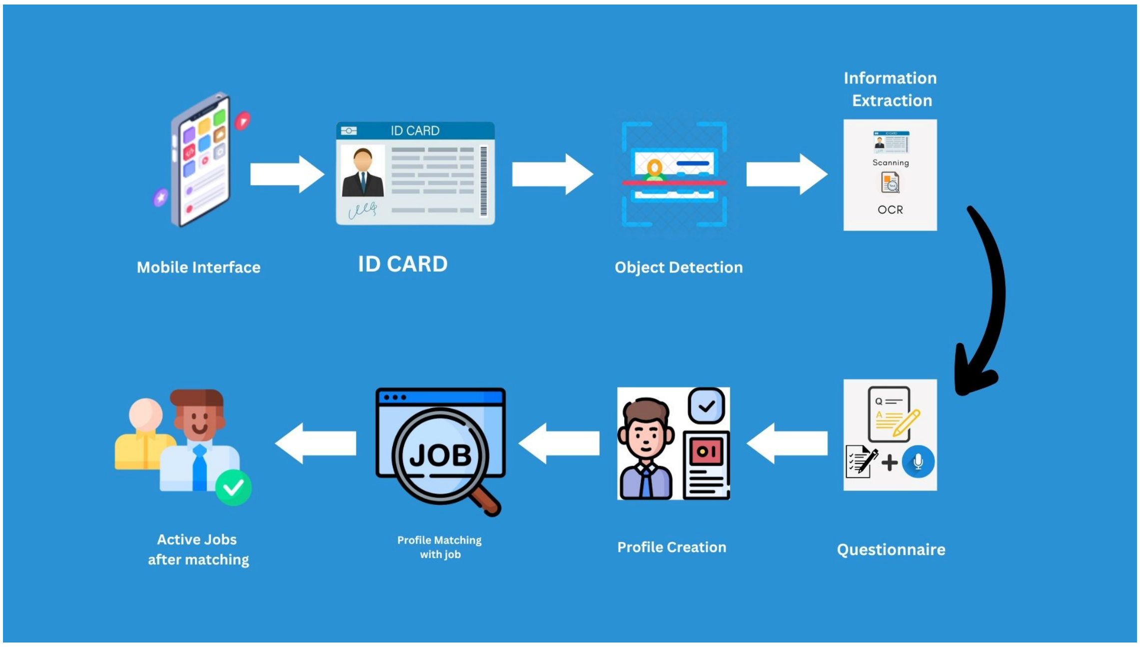

Figure 4.2: Activity Diagram

• **Class Diagram**
This is the class diagram of the Skill Bazaar application. In which there are the main classes. It can be seen how each of the class is connected to each other and their dependencies as well. The tentative data types of the classes are also described as for the project we are going to use a database to store the data.

1. User
   (a) Worker
      i. Blue collar worker
      ii. Skillful person
   (b) Business
      i. Individual
      ii. Company
2. Jobs/Services
3. Bookings
4. Ratings

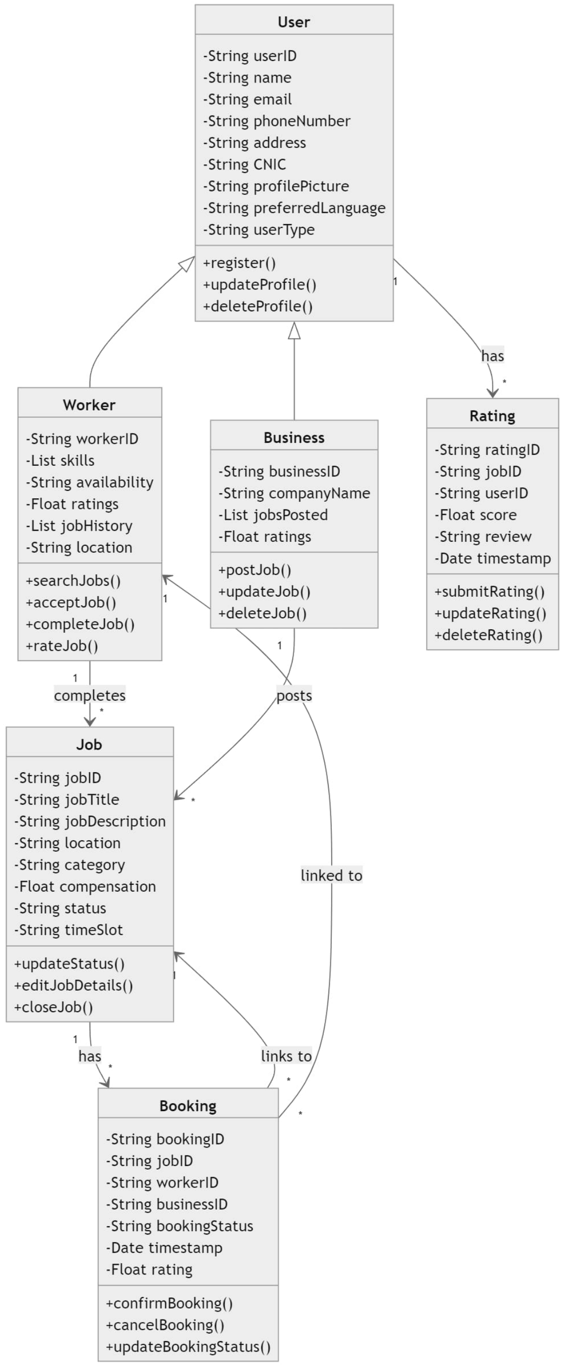

Figure 4.3: Class Diagram

• **Class-level Sequence Diagram**
In the class level sequence Diagram, we can see the users as workers or business. They both interact with each other in terms of; Here are some states which the user would be in;

1. Job Selection
2. Job posting
3. Job Booking
4. Communication
5. Payment

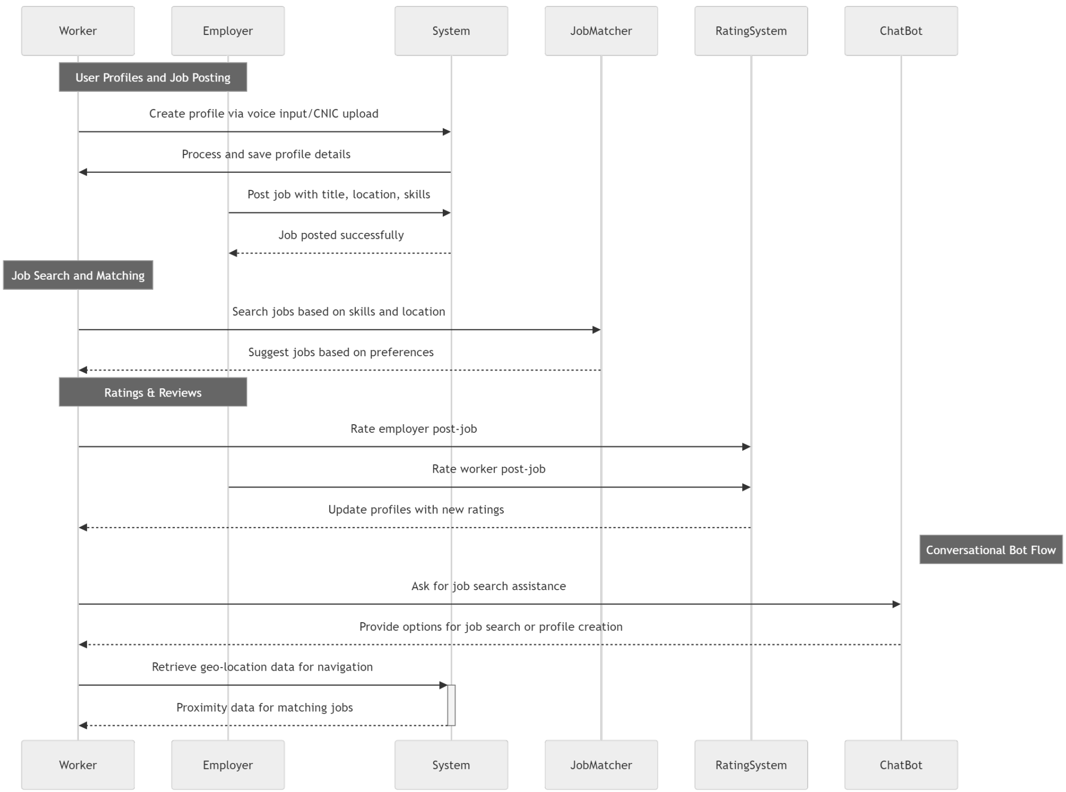

Figure 4.4: Class Level Sequence Diagram

• **State Transition Diagram** (for the projects which include event handling and back-end processes)
Here are some states which the user would be in;

1. Initial State (Idle)
2. Profile Creation State

3. Job Posting State

4. Job Search State

5. Job Matching & Recommendation State

6. Voice Interaction State

7. Error Handling State

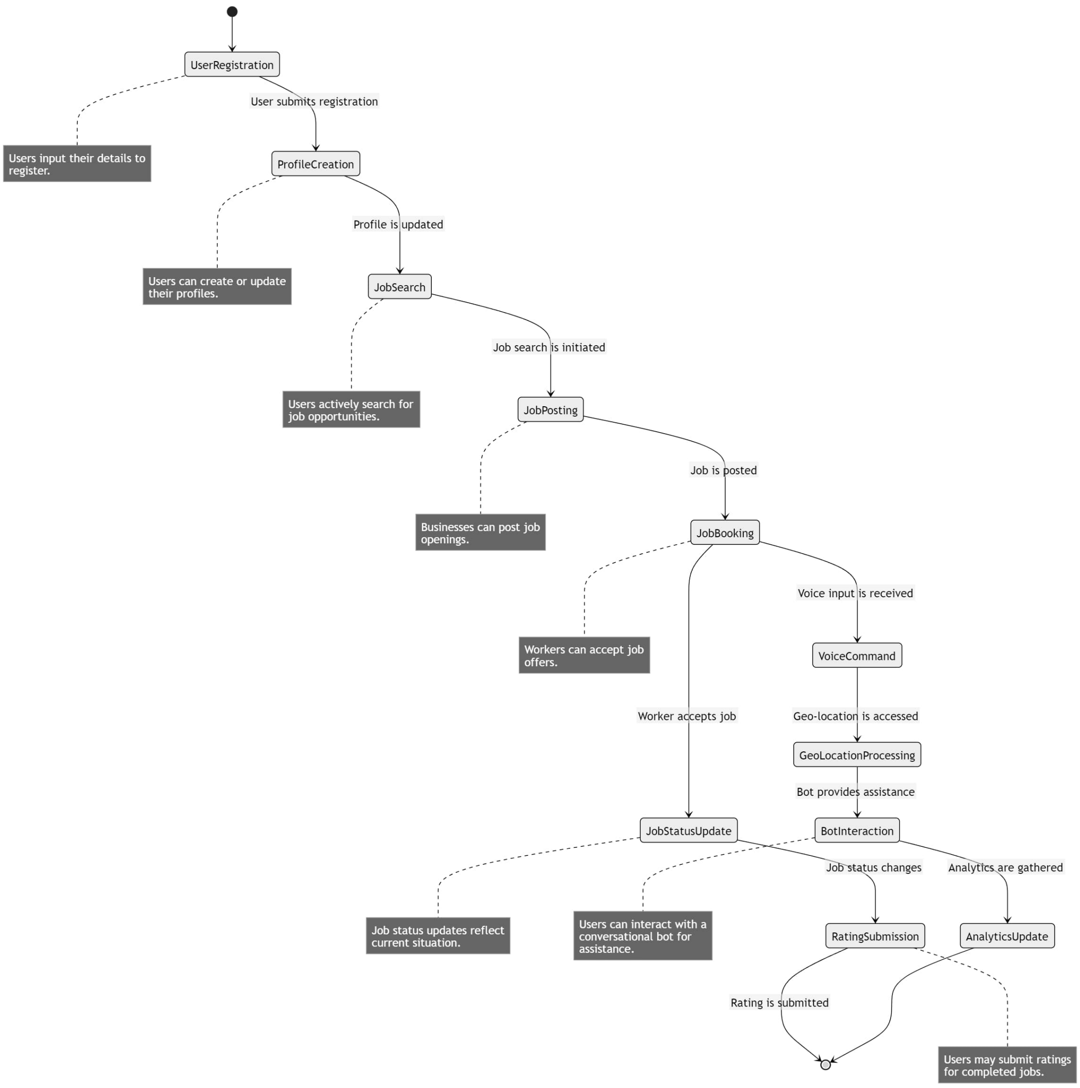

Figure 4.5: State Transition Diagram

Design Models for Procedural Approach

The applicable models for the project using procedural approach may include:
• **Activity Diagram**

Here is the activity diagram for the procedural Approach. it can be seen that the user is asked to take a picture of his/her CNIC and then OCR will be used for information retrieval and the user can input using both text which will be processed and information will be retrieved from it and also using ASR technology our application will allow voice input as well. Once all of the necessary information is collected from the user, the application will automatically generate the profile for the user, job posting and searching will also voice assisted and there is also a voice based conversational bot for user's ease which will help them in successful job matching that is done using AI matching algorithm. Our system will generate analytics and provide relevant responses to improve user experience. The database is managed and the data is saved as well.

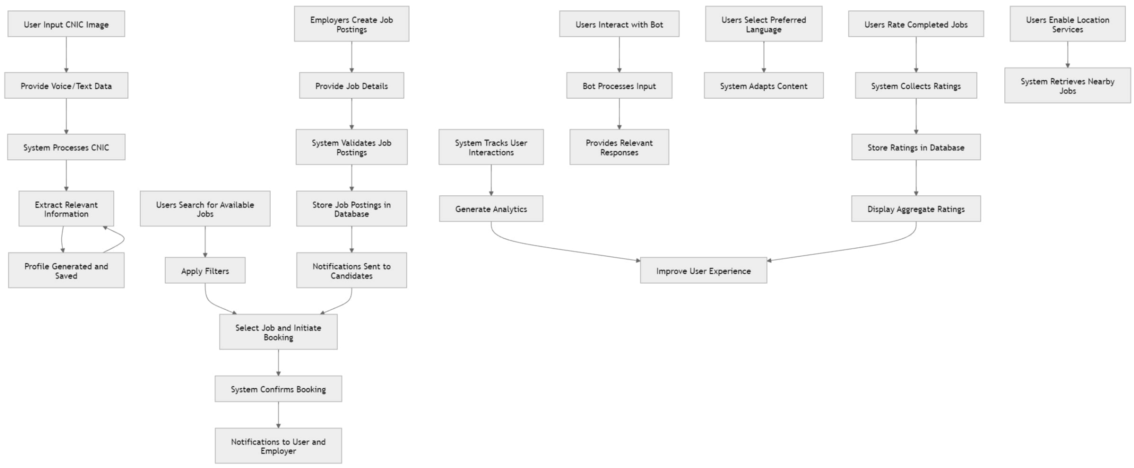

Figure 4.6: Simple Activity Diagram

Now, below are the step by step activity diagrams for each of the procedures.

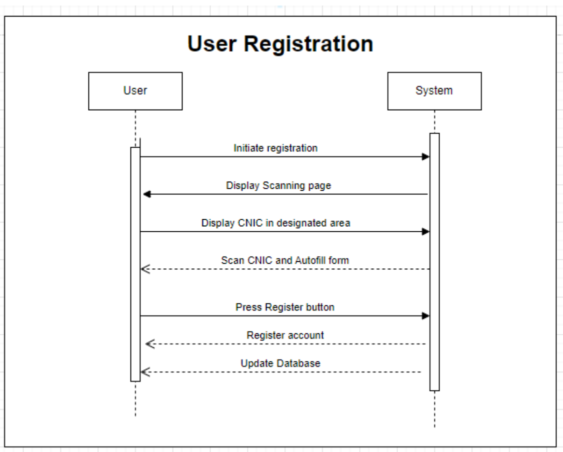

Figure 4.7: User Registration Diagram

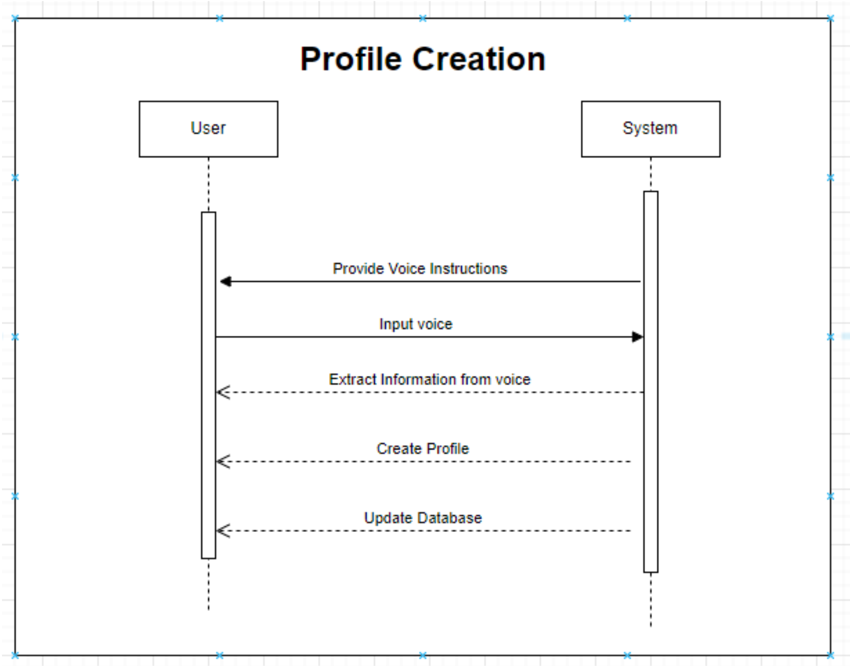

Figure 4.8: Profile Creation Diagram

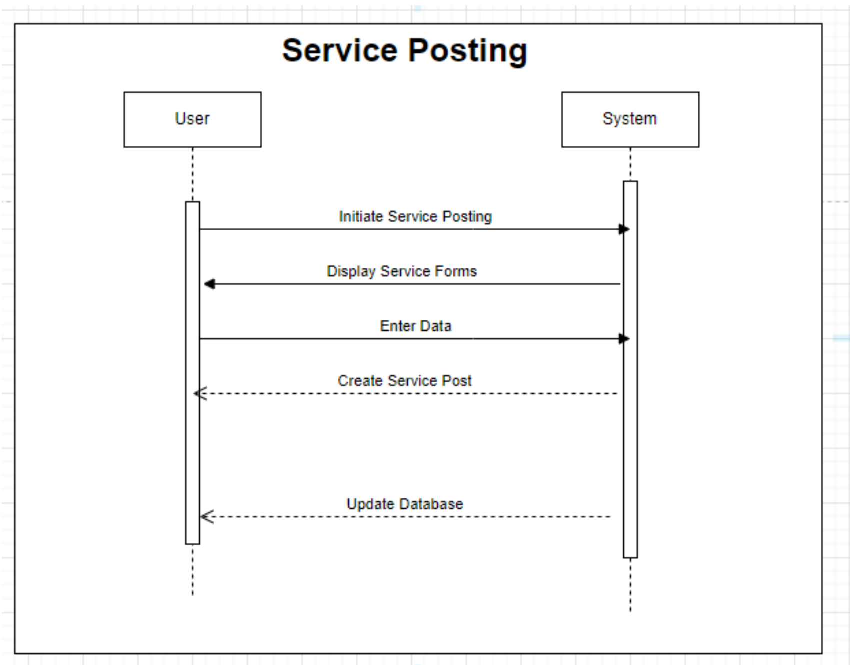

Figure 4.9: Service/ Job Posting

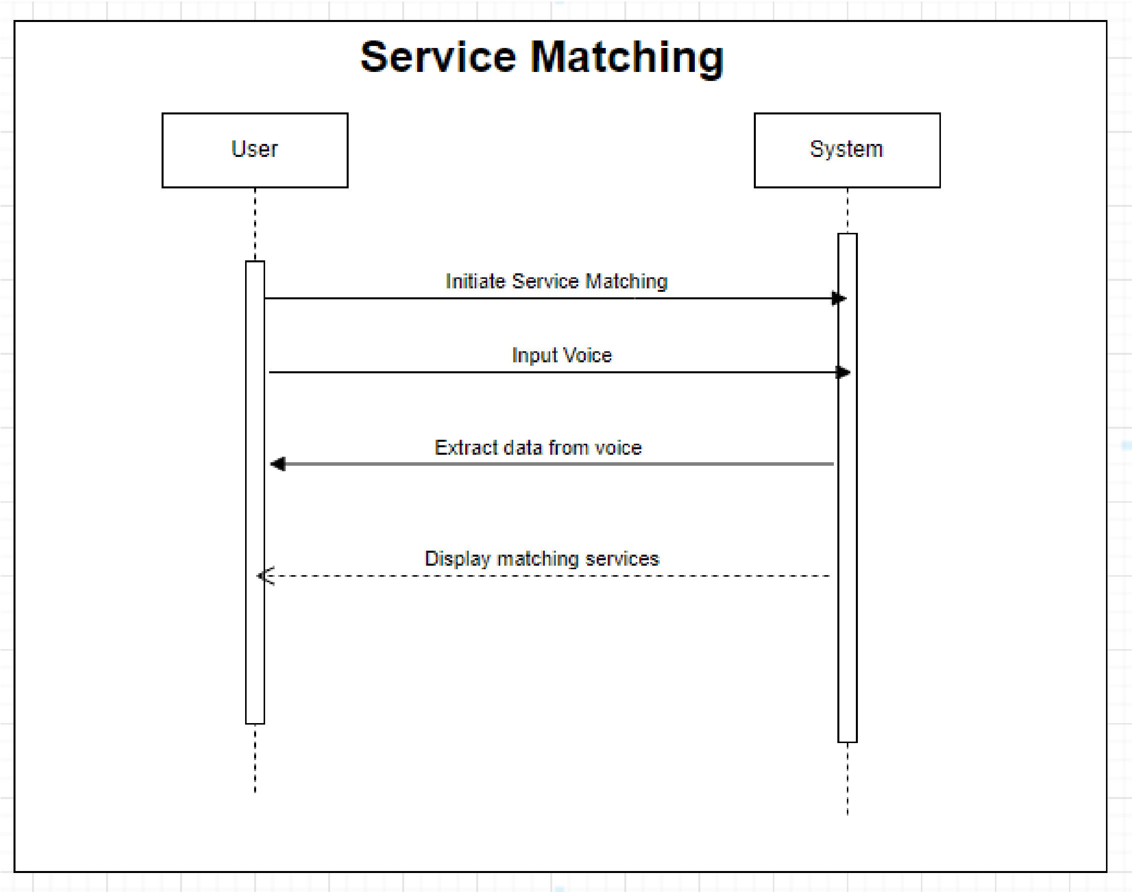

Figure 4.10: Service/Job Matching

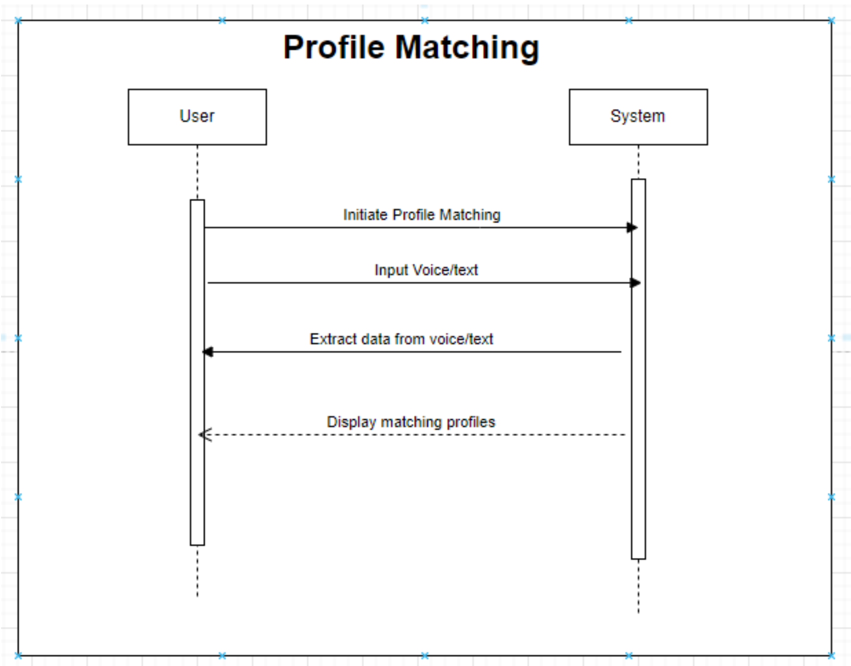

Figure 4.11: Profile Matching Diagram

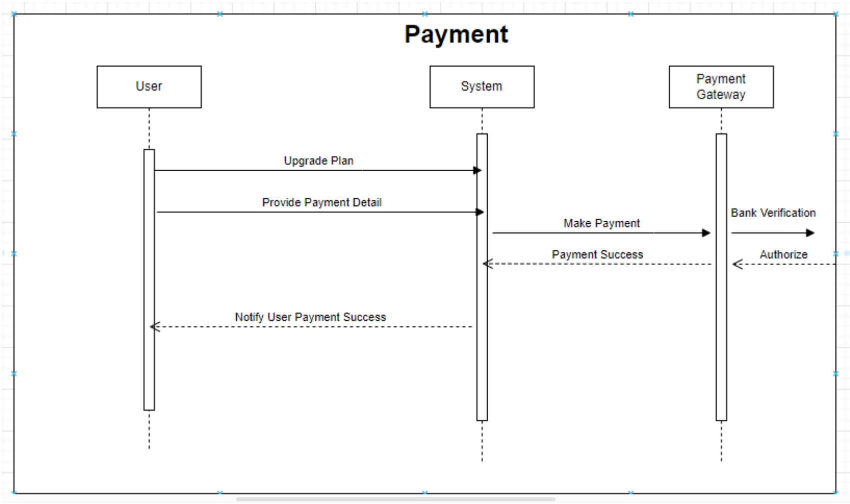

Figure 4.12: Payment Diagram

• Data Flow Diagram

This is the simple data flow diagram for the application, so there are basically four types of databases that are being maintained.

1. User, Job
2. Booking, Ratings

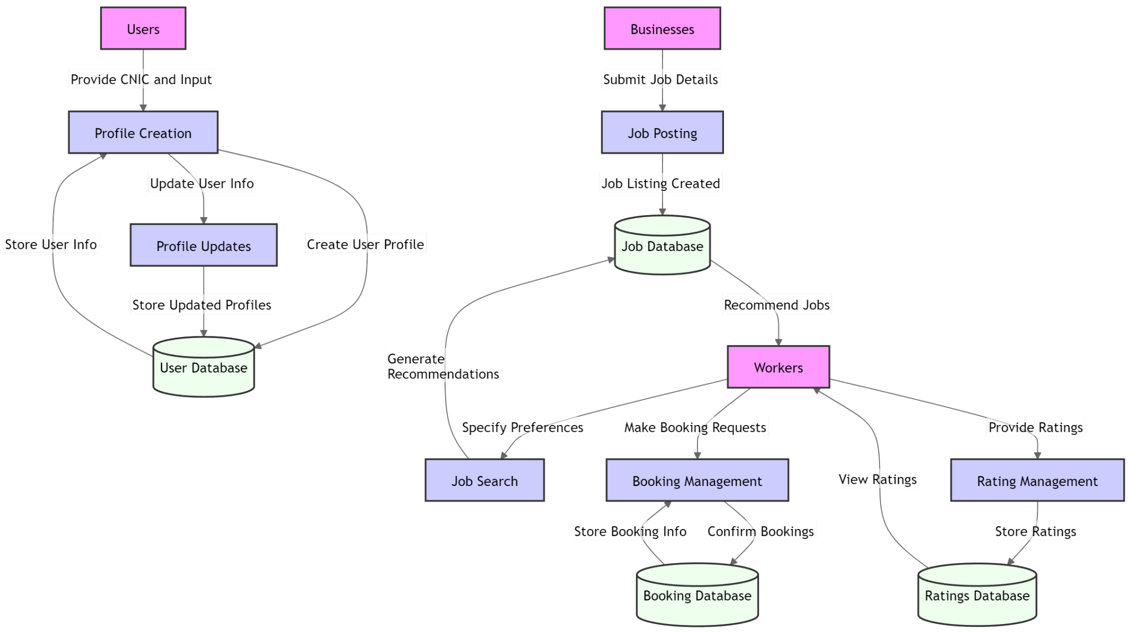

Figure 4.13: Data Flow Diagram

- **System Architecture Diagram**

This is the simple system architecture diagram for the application.

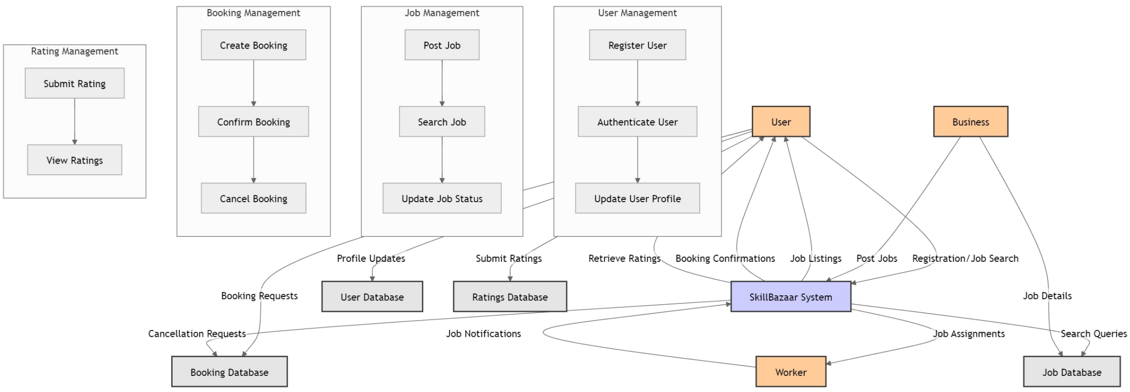

Figure 4.14: System Architecture Diagram

- **State Transition Diagram**

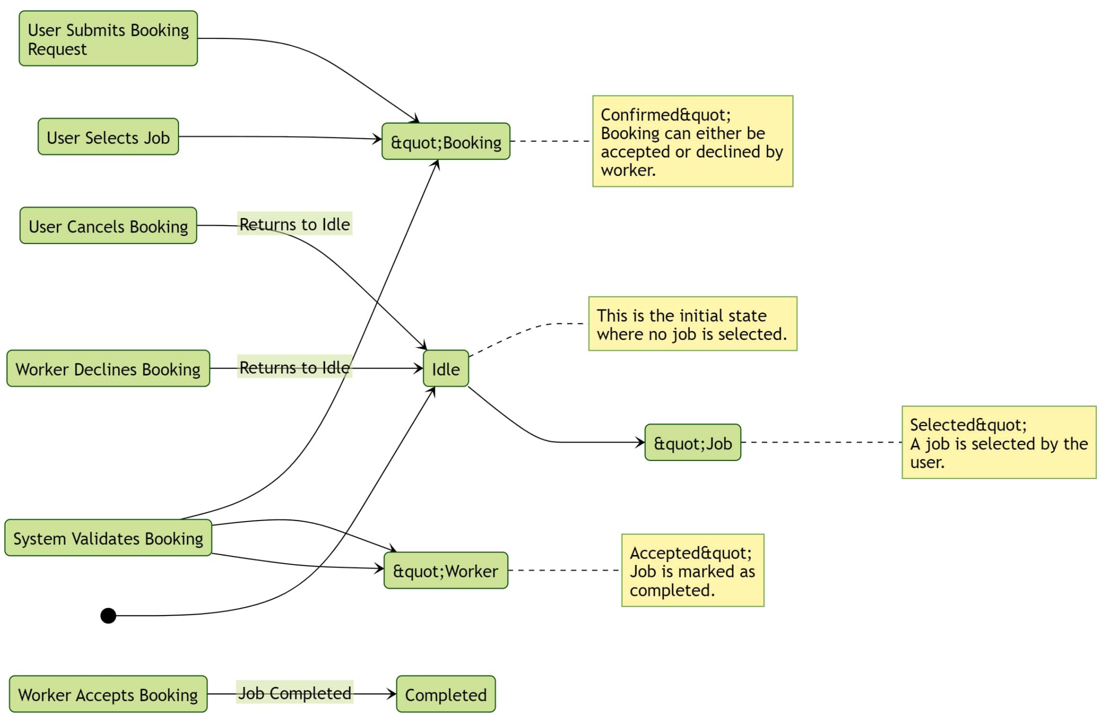

Figure 4.15: State Transition Diagram

- **System-level Sequence Diagram**

This is the system level sequence Diagram

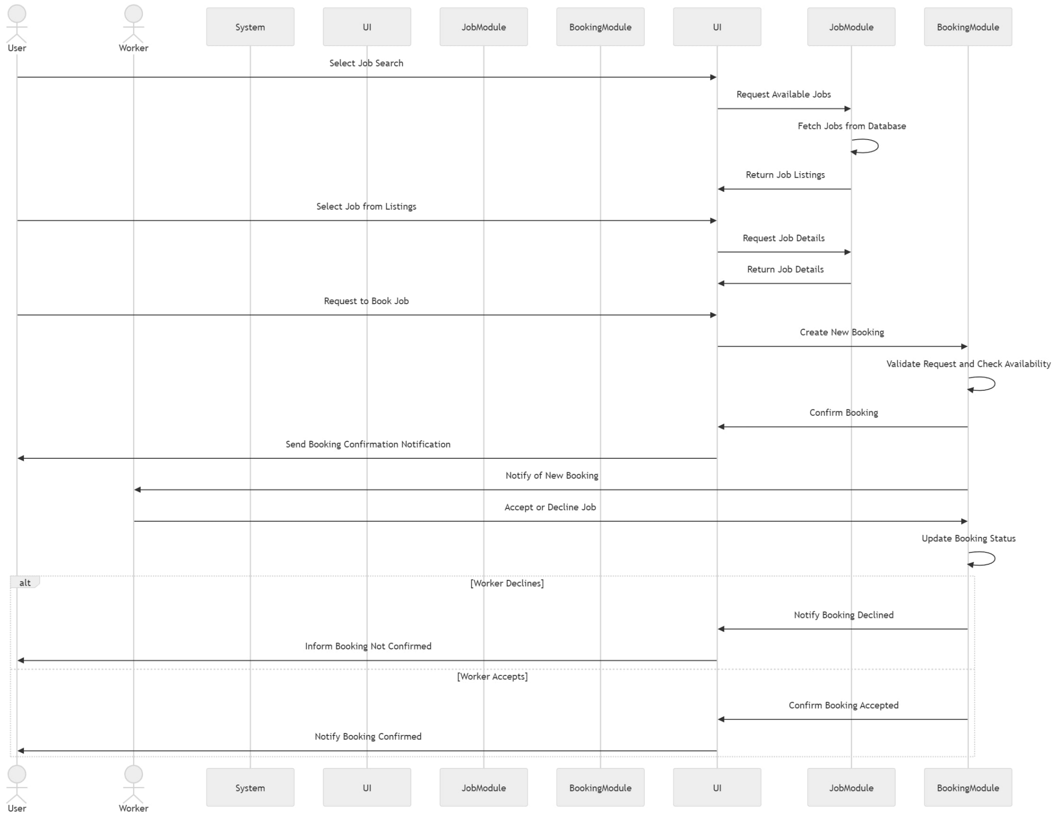

Figure 4.16: System Sequence Diagram

4.3 Data Design

Information Domain Transformation

In the SkillBazaar project, the information domain is transformed into structured data to facilitate efficient storage, retrieval, and processing of user interactions, job postings, and other essential functionalities. The primary focus is on organizing data into entities that represent key aspects of the system, ensuring that data integrity and accessibility are maintained.

Major Data Entities

1. **User**:

    - **Attributes**: userID, name, email, phoneNumber, address, CNIC, profilePicture, preferredLanguage, userType.
    - **Storage**: User data is stored in a relational database table named `Users`, where each row corresponds to a user account, and attributes are represented as columns. This allows for efficient queries based on user information.

2. **Worker** (Subclass of User):

    - **Attributes**: workerID, skills, availability, ratings, jobHistory, location.
    - **Storage**: Worker-specific data is stored in a table called `Workers`, which references the `Users` table through the userID. This structure supports easy retrieval of worker details while maintaining normalization.

3. **Business** (Subclass of User):

    - **Attributes**: businessID, companyName, jobsPosted, ratings.
    - **Storage**: Business data is stored in a `Businesses` table, which also references the `Users` table. This allows businesses to be treated as a specific type of user while retaining all relevant information.

4. **Job**:

    - **Attributes**: jobID, jobTitle, jobDescription, location, category, compensation, status, timeSlot.
    - **Storage**: Job listings are organized in a `Jobs` table, with each job entry including the corresponding workerID and businessID, allowing for easy associations with users and tracking of job postings.

5. **Booking**:

    - **Attributes**: bookingID, jobID, workerID, businessID, bookingStatus, timestamp, rating.
    - **Storage**: Booking records are kept in a `Bookings` table, linking to the `Jobs` table to identify which job has been booked, as well as to the `Workers` and `Businesses` tables for complete context.

6. **Rating**:

    - **Attributes**: ratingID, jobID, userID, score, review, timestamp.
    - **Storage**: Ratings are stored in a `Ratings` table that references both the `Jobs` and `Users` tables, allowing users to provide feedback on jobs effectively.

7. **Voice Input**:

    - **Attributes**: voiceID, userID, voiceData, recognizedText, inputType.
    - **Storage**: Voice inputs are stored in a `VoiceInputs` table, capturing recorded data along with metadata about the user and the type of input.

8. **GeoLocation**:

- **Attributes**: geoLocationID, latitude, longitude, radius.
- **Storage**: Geo-location data is kept in a `GeoLocations` table, which is used for matching workers to jobs based on their proximity.

9. **Conversational Bot**:

- **Attributes**: botSessionID, userID, interactionHistory, currentIntent.
- **Storage**: Bot interactions are logged in a `BotSessions` table, maintaining records of user interactions with the conversational agent.

10. **Analytics**:

- **Attributes**: analyticsID, userID, jobCompletionRate, userEngagement, averageRating.
- **Storage**: Analytics data is stored in an `Analytics` table, aggregating performance metrics for users over time.

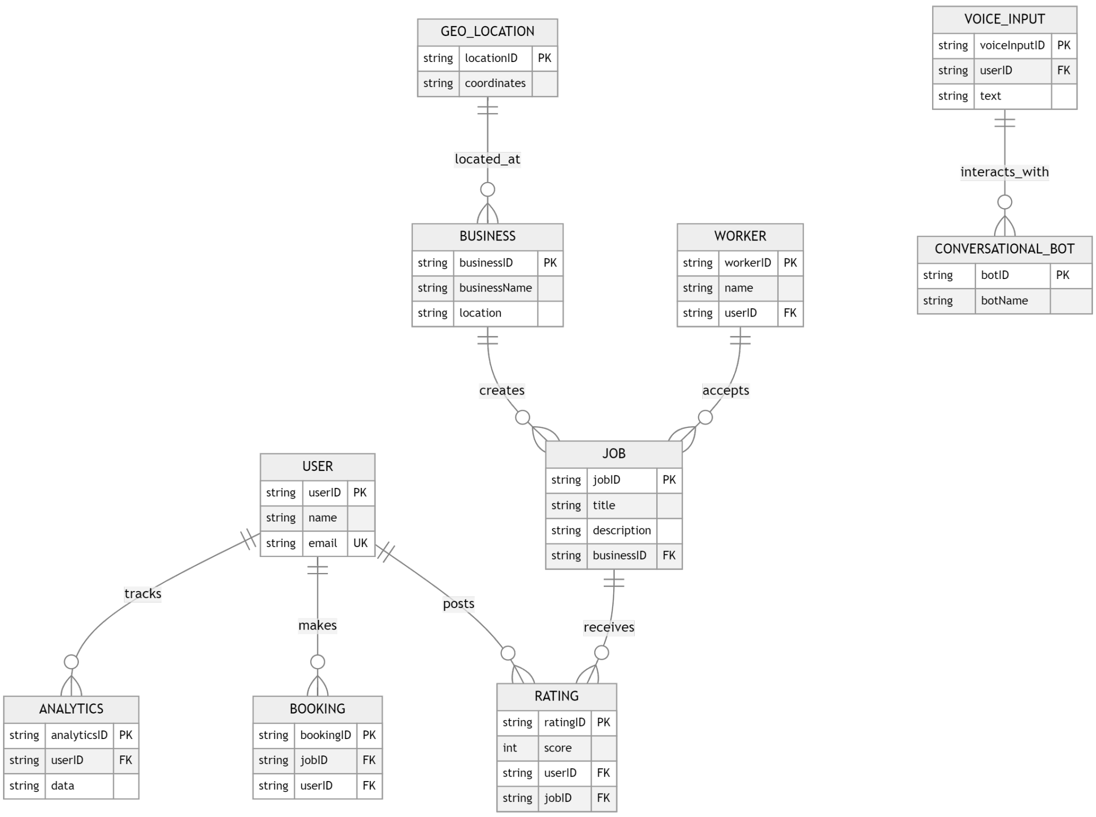

Figure 4.17: Data Design Diagram

Data Storage The primary data storage for the SkillBazaar project utilizes a relational database management system (RDBMS), such as PostgreSQL or MySQL, which provides robust support for structured data and complex queries. The database schema is designed to minimize redundancy and ensure data integrity through the use of foreign keys and relationships among tables.

Additionally, auxiliary storage solutions may be employed for specific use cases:

- **File Storage**: For user-uploaded profile pictures, recorded voice data, and other binary files, a cloud storage solution (e.g., Amazon S3 or Google Cloud Storage) may be utilized.
- **Caching**: To improve performance and reduce database load, caching mechanisms (e.g., Redis or Memcached) can be implemented for frequently accessed data, such as job listings or user profiles.

Chapter 5

Figma Design

5.1 Story Boarding

Below is the figma design for the 6 pages that link directly to our use cases and mid evaluation as we would be working on them. The interface is easy to understand as our most users are most likely less educated who are not good at english or urdu so with their minimal input, we will automate their profiles for them.

Figure 5.1: Registration

Figure 5.2: CNIC Image

Figure 5.3: Registration Form

Figure 5.4: Job seeker's Home Page

Figure 5.5: Profile Creation

Figure 5.6: Complete Profile

Chapter 6

Conclusions and Future Work

All of us have seen daily wage workers sitting beside roads in harsh weather conditions looking for work everyday and there are some skilled women/men who can do embroidery, handicraft and pottery can not sell their products without the help of entrepreneurs. They all suffer in poverty because there is no bridge that can connect the employers with the skilled laborers or blue collar workers irrespective of the language and educational barriers so our motive is to build such application that can help them find job opportunities easily and conveniently so we want to provide them a platform where they can find jobs easily, sell their services and earn their due share more conveniently. Skill Bazaar is an application that automates their profiles for them, allows voice input and a user friendly conversational bot which will resolve their queries.

Future works:

1. Pakistan is a diverse country in which there are 4 major languages along with their dialects and most of them are not able to understand Urdu so our application will do Regional Languages Integration so that people from different backgrounds can get job opportunities too.

2. People can automate their profile even if they do not have smart phone (as many blue collar job workers are living hand-to-mouth life) by communicating with our customer support and each time their profile finds a good job opportunity, the application generates a call to them informing them about the job opportunities.

Bibliography